PHYS-C0253 - Quantum Mechanics

Exercise set 1

Due date: April 28, 2026 by 23:59 on MyCourses
Return the exercises as a `.pdf`.
You can write by hand and take pictures, use digital note-taking or LaTeX etc.

1. Consider the vectors $|\psi\rangle = 4|\phi_1\rangle + i|\phi_2\rangle$ and $|\chi\rangle = 2|\phi_1\rangle + (1-4i)|\phi_2\rangle$, where $|\phi_1\rangle$ and $|\phi_2\rangle$ are orthonormal, i.e. $\langle\phi_k|\phi_m\rangle = \delta_{km}$, where $\delta_{km} = 1$ for $k = m$ and $\delta_{km} = 0$ for $k \neq m$.

   (a) Express $|\psi\rangle + |\chi\rangle$ and $\langle\psi| + \langle\chi|$ in their simplest form using $|\phi_1\rangle$ and $|\phi_2\rangle$.

   (b) Express $|\phi_1\rangle$ in terms of $|\psi\rangle$ and $|\chi\rangle$.

   (c) Calculate the inner products $\langle\psi|\chi\rangle$ and $\langle\chi|\psi\rangle$. Are they equal?

   (d) Show that $|\psi\rangle$ and $|\chi\rangle$ satisfy the Cauchy–Schwarz inequality and the triangle inequality.

2. Consider the so-called Pauli operators $\hat{\sigma}_x = |0\rangle\langle 1| + |1\rangle\langle 0|$, $\hat{\sigma}_y = -\mathrm{i}|0\rangle\langle 1| + \mathrm{i}|1\rangle\langle 0|$ and $\hat{\sigma}_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, where $\{|0\rangle, |1\rangle\}$ form an orthonormal basis of the considered Hilbert space.

   (a) Show that each Pauli operator is Hermitian.

   (b) Write down the matrix representation of the Pauli operators. Hint: for an operator $\hat{A}$ and an orthonormal basis $\{|\phi_k\rangle\}_k$, the matrix element A_{jk} is defined as $\langle\phi_j|\hat{A}|\phi_k\rangle$.

   (c) Solve the eigenvalues and the corresponding eigenstates of each Pauli operator using the matrix form, and write the eigenstates using the ket vectors $|0\rangle$ and $|1\rangle$.

   (d) For each Pauli operator, show that the eigenstates are orthogonal.

3. (a) Prove the Cauchy–Schwarz inequality $|\langle\psi|\phi\rangle| \leq ||\psi||\,||\phi||$. Here we use the shorthand notation $||\psi||$ $(= ||\,|\psi\rangle||)$ for the norm of $|\psi\rangle$ as on lectures. Hint: Start from $0 \leq ||\,|\psi\rangle + \lambda|\phi\rangle\,||$ and choose the scalar $\lambda \propto \langle\phi|\psi\rangle$ in a clever way.

   (b) Prove the triangle inequality $||\,|\psi\rangle + |\phi\rangle|| \leq ||\psi|| + ||\phi||$. Hint: Calculate $||\,|\psi\rangle + |\phi\rangle\,||^2$ and use (a). You may also use the fact that $\mathrm{Re}(z) \leq |z|$ for a complex number z.

(c) Demonstrate the necessary and sufficient conditions for these inequalities to become equalities. Hint: Let $a\,|\psi\rangle = \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle}\,|\psi\rangle$ be the *projection* of $|\phi\rangle$ on to $|\psi\rangle$. You can write $|\phi\rangle$ in terms of the projection and the *rejection* $|\chi\rangle = |\phi\rangle - \frac{\langle\psi|\phi\rangle}{\langle\psi|\psi\rangle}\,|\psi\rangle$ as $|\phi\rangle = a\,|\psi\rangle + |\chi\rangle$. Note that the rejection is orthogonal to $|\psi\rangle$, i.e. $\langle\chi|\psi\rangle = 0$.

4. In quantum mechanics, we often cannot solve the eigenvalue problem $\hat{H}\,|\psi\rangle = E\,|\psi\rangle$ exactly. The Rayleigh quotient $R(\psi) = \frac{\langle\psi|\hat{A}|\psi\rangle}{\langle\psi|\psi\rangle}$ provides a powerful tool for estimating the lowest eigenvalue a_0, without knowing the exact eigenstates.

   (a) Prove that if $\hat{A}$ is a Hermitian operator, then:

   (i) All eigenvalues a_n are real.

   (ii) Eigenvectors $|n\rangle$ and $|m\rangle$ corresponding to distinct eigenvalues ($a_n \neq a_m$) are orthogonal.

   (b) Let $\hat{A}$ be a Hermitian operator on an infinite-dimensional Hilbert space with a discrete, non-degenerate spectrum: $\hat{A}\,|n\rangle = a_n\,|n\rangle$ for $n = 0, 1, 2, \ldots$. We order the eigenvalues such that $a_0 < a_1 < a_2, \ldots$. Let $|\psi\rangle$ be an arbitrary state vector in the infinite-dimensional Hilbert space. Prove that then

$$R(\psi) \geq a_0.$$

   When does the equality hold?

   Hint: Expanding the abstract vector $|\psi\rangle$ might be helpful. The eigenbasis of $\hat{A}$ forms a complete basis on the (infinite-dimensional) Hilbert space in this case.

   (c) Suppose the smallest eigenvalue a_0 is g-fold degenerate. That means there are g orthonormal eigenvectors $|0_1\rangle, |0_2\rangle, \ldots, |0_g\rangle$ all with eigenvalue a_0. Would the inequality $R(\psi) \geq a_0$ still hold? When does the equality hold in this case? Briefly explain.

   (d) Assume that we know the exact eigenvector $|0\rangle$ of $\hat{A}$. We choose a normalized state $|\phi\rangle$ (i.e., $\langle\phi|\phi\rangle = 1$) that is orthogonal to the ground state, i.e., $\langle 0|\phi\rangle = 0$. Prove that

$$\langle\phi|\hat{A}|\phi\rangle \geq a_1.$$